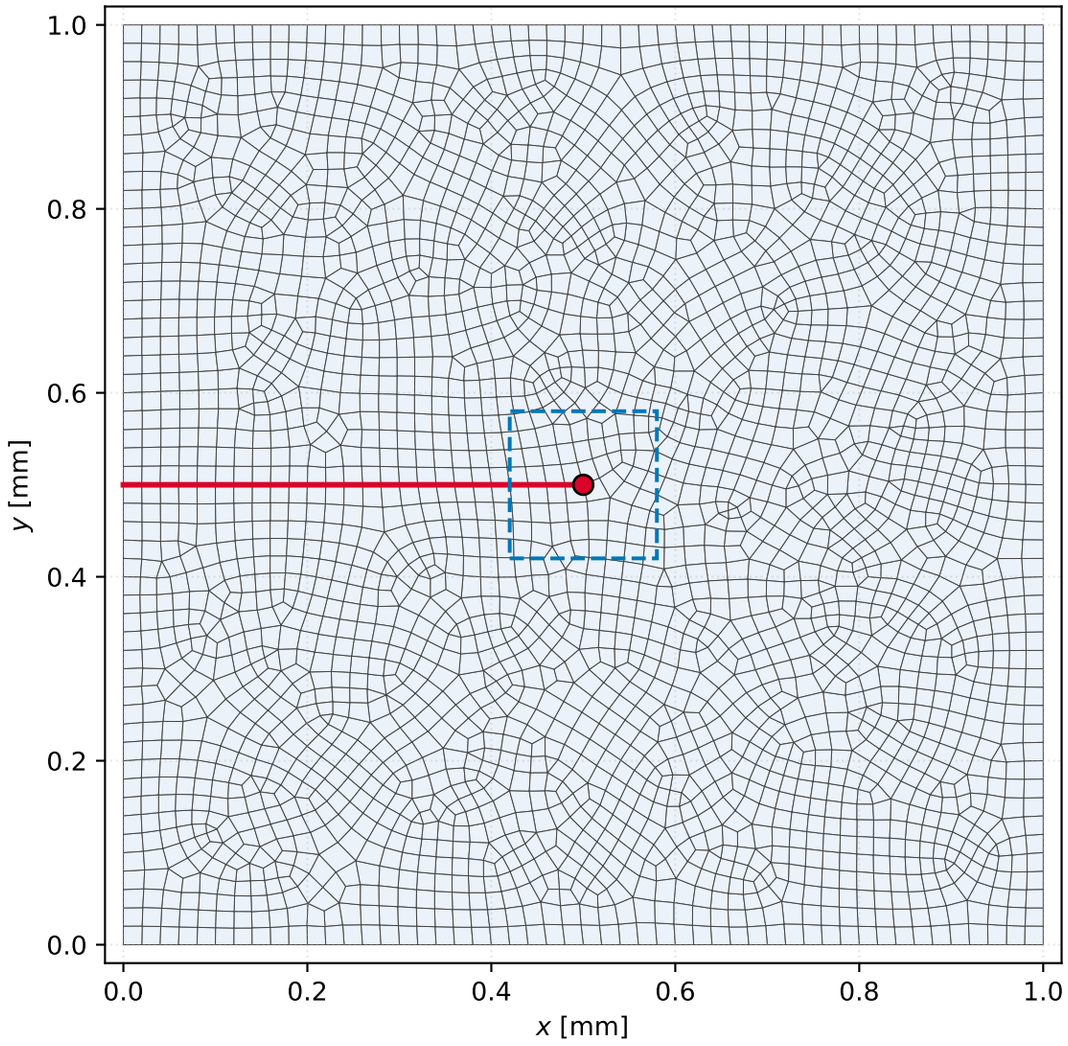
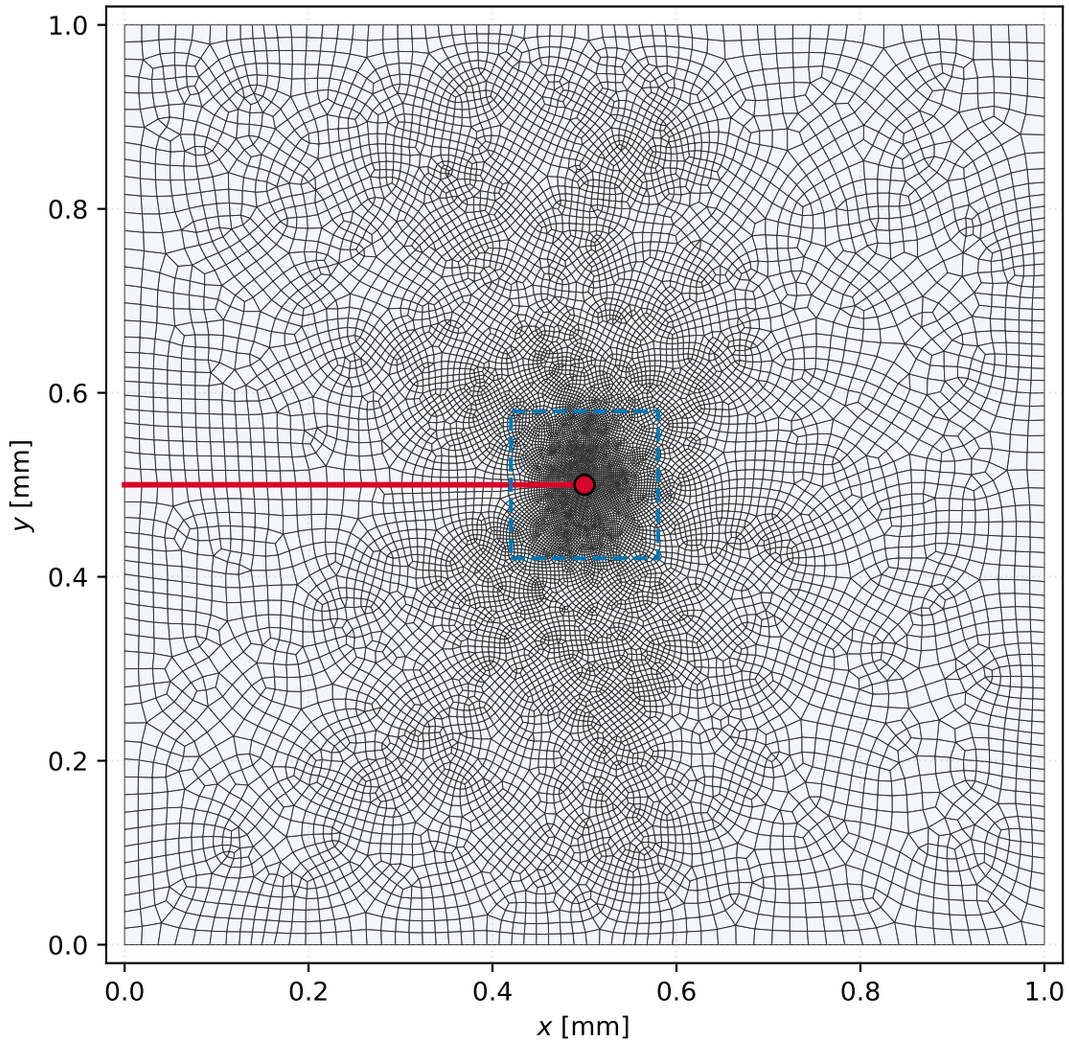
**Discretization Resolution Comparison: 2,906-Element Coarse Baseline vs. 14,677-Element Method B Adapted Mesh (Local Tip Resolution Enhancement: 10.1x Finer Sizing, 101x Area Density Contrast)**

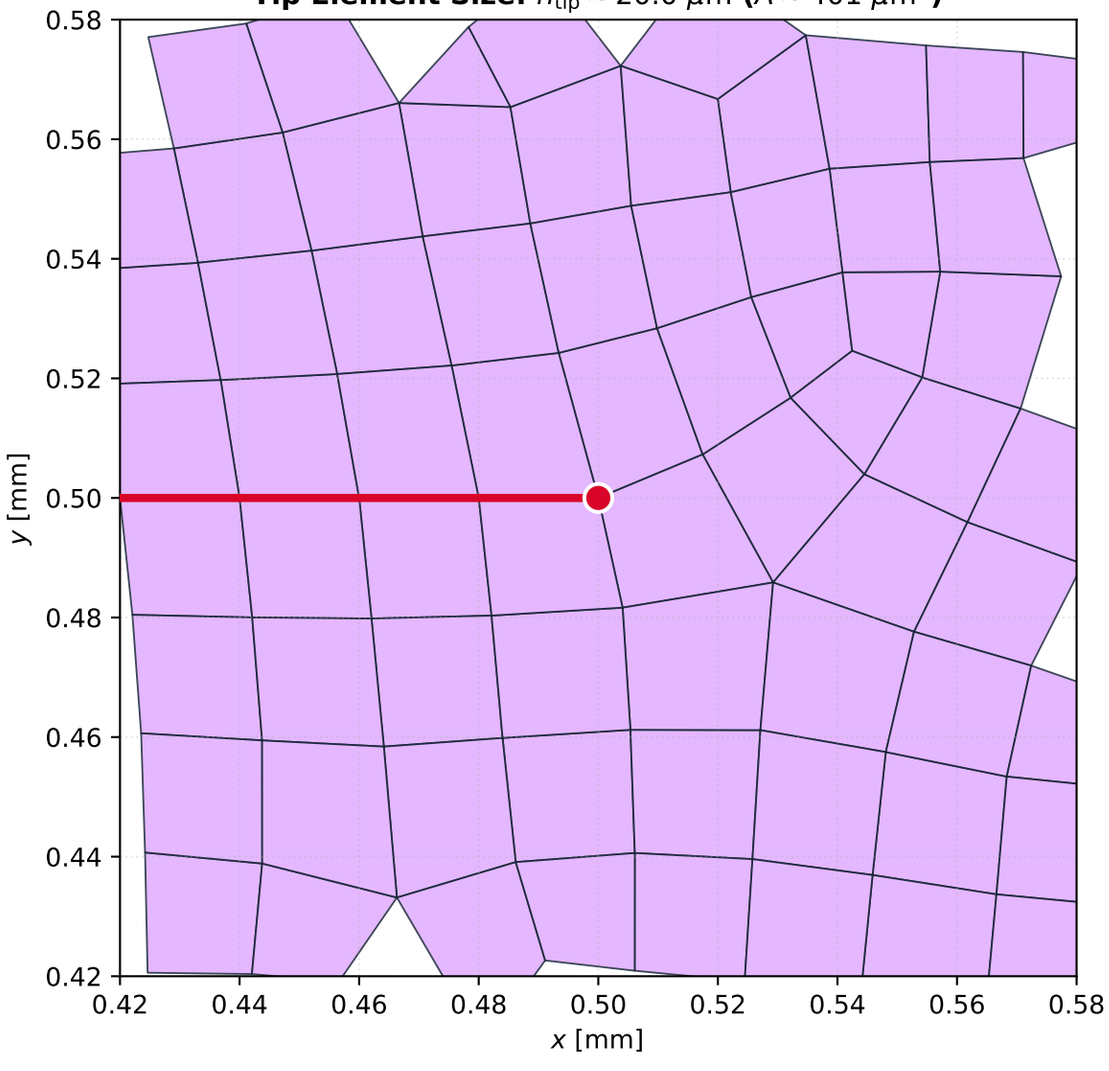
**(a) Coarse Pre-Analysis Mesh: Full Domain**
**2,906 Elements (2,988 Nodes)**




**(b) Method B Native Adapted Mesh: Full Domain**
**14,677 Elements (14,646 Nodes)**




**(c) Coarse Mesh: Crack-Tip Region**
**Tip Element Size:** $h_{tip} \approx 20.0\ \mu m$ **(**$A \approx 401\ \mu m^2$**)**




**(d) Method B Adapted: Crack-Tip Region**
**Tip Element Size:** $h_{tip} \approx 1.99\ \mu m$ **(**$A \approx 3.9\ \mu m^2$**)**

